

Appendix A

Protection Terminology

There are many terms and definitions used in power system protection that must be clearly understood prior to any detailed discussion of the subject. This appendix presents some of the terms that are required for a preliminary understanding. Other terms are defined throughout the text.

Terms and definitions related to the general subject of power system protection are presented first. This is followed by terms used in relaying and a classification of relays used in the industry. Finally, circuit breaker terms and definitions are discussed.

A.1 Protection Terms and Definitions

There are many terms used in the publications and discussions associated with power system protection. We present here some of the basic terms and provide formal definitions of each.

Application of protection systems. Protection systems are applied to utility, industrial, and commercial electrical power systems in all their critical sectors such as generation, transmission, and distribution. The main function of the protection system is to sense and identify power system faults when occur and isolate the faulted parts of the system as quickly as possible. The intent is to avoid excessive damage and to limit the outage to the affected parts. Protection systems are required to meet high standards of reliability, speed, selectivity, and economics [1, 2].

Reliability of protection systems. Reliability reflects the assurance of correct relay operation (dependability) and against incorrect operation (security).

The following definitions are associated generally with power system protection. These definitions have been presented formally in the IEEE standards related to nuclear power generating stations. Several of these general definitions form an excellent basis for power systems in general. These definitions follow:

Protection system. The electrical and mechanical devices (from measured process variables to protective action system input terminals) involved in generating those signals associated with the protective functions [3].

Protective function. The sensing of one or more variables associated with a particular generating station condition, the signal processing, and the initiation and completion of the protective action at values of the variables established in the design bases [3–5].

Protective action. The initiation of the operation of a sufficient number of actuators to effect a protective function [3–7].

In some complex process industries, such as electric power generating plants, there are many systems that require protective system surveillance and action. Examples include the plant electric systems, hydraulic systems, pneumatic systems, safety systems, and others. Electric power transmission and distribution system protection deals almost exclusively with electric system protection. This is also an important aspect of electric power generating stations. This book concentrates on the protection of electric systems, although many of the principles can be applied to other problems.

The protective function, in electric system protection, is provided by *relays*, which are defined as follows:

Relay (general). An electric device that is designed to respond to input conditions in a prescribed manner and, after specified conditions are met, to cause contact operation or similar abrupt change in associated electric control circuits.

Note A. Inputs are usually electrical, but may be mechanical, thermal, or other quantities. Limit switches and similar simple devices are not relays.

Note B. A relay may consist of several units, each responsive to a specified input, the combination of units providing the overall desired performance characteristic of the relay [3, 8–10]. One type of relay with several units is a multifunction relay. This type is a single physical unit that includes multiple functions usually processing multiple inputs. The multifunction relay offers several advantages, such as speed, compactness, minimum wiring, economy, and versatility. There are also disadvantages such as risk of common mode failure of the software or hardware platform [11].

The actuator that provides the protective action, in an electric system, is usually a device called a circuit breaker, which is defined as follows:

Circuit breaker (general). A mechanical switching device capable of making, carrying, and breaking currents under normal circuit conditions, and also, making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions, such as those of short circuit [3, 12, 13].

Note 1. A circuit breaker is usually intended to operate infrequently, although some types are suitable for frequent operation.

Note 2. The medium in which circuit interruption is performed may be designated by a suitable prefix, such as, air-blast circuit breaker, air circuit breaker, compressed-air circuit breaker, gas circuit breaker, oil circuit breaker, vacuum circuit breaker, etc.

Note 3. Circuit breakers are classified according to their application or characteristics and these classifications are designated by the following modifying words or clauses delineating the several fields of application, or pertinent characteristics:

High-voltage power	Rated 1.0 kV ac and above
Low-voltage power	Rated below 1.0 kV
Other (see ANSI standards [8])	

Finally, we introduce a general term that is often used in system protection.

Switchgear. A general term covering switching and interrupting devices and their combination with associated control, instrumentation, metering, protective, and regulating devices, also assemblies of these devices with associated interconnections, accessories, and supporting structures

used primarily in connection with the generation, transmission, distribution, and conversion of electric power [3].

Switchgear is a broad term that covers all of the hardware usually associated with the protective system, including both function and actuation.

A.2 Relay Terms and Definitions

In addition to the protective system definitions, it is important for the student of system protection to be familiar with definitions used in connection with the relays themselves. These definitions are, for the most part, taken from IEEE Standard 313 [10], which contains many definitions associated with relaying and protective systems in general. We cite here only those that are considered necessary as a beginning. Other definitions are introduced throughout the book as needed.

Auxiliary relay. A relay whose function is to assist another relay or control device in performing a general function by supplying supplemental actions.

Relay logic is often designed with more than one relay required to achieve the desired action. Auxiliary relays are often used to provide the necessary supplementary action, such as providing additional current-carrying capacity, providing a seal-in function, adding time delay, or providing some kind of interlock.

Burden. The load impedance imposed by a relay on an input circuit, expressed in ohms and phase angle at a specified condition.

The relay represents an electrical load to the input device; for example, an overcurrent relay has its input supplied by a current transformer. The input device has a rating that states its capability of providing the required quantity. The relay burden gives the impedance of the relay, such that the rating of the input device can be determined.

Contact. A conducting part that acts with another conducting part to make or break a circuit.

Back contact (“b” contact). A contact that is closed when the relay is reset.

Front contact (“a” contact). A contact that is closed when the relay is picked up.

Contact opening time. The time a contact remains closed while in process of opening following a specified change of input.

Contacts are depicted in relay circuit diagrams as shown in Figure A.1. Note that the “a” contact is always shown in drawings with the contact open, even though the contacts may be normally closed during operation. Similarly, the “b” contact is always shown as normally closed, even though its normal operating state may occur with the input quantity having sufficient magnitude to hold the contact open.

We usually think of a contact as one of the precious metals (gold or silver) that is caused to connect or “make” with a second contact to complete a circuit mechanically. Actually, this can also be accomplished electronically, for example, by means of a thyristor.

Dropout. A term for contact operation (opening or closing) as a relay just departs from pickup. Also, the maximum value of an input quantity that will allow a relay to depart from pickup.

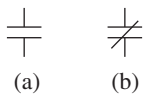


Figure A.1 ANSI standard graphic symbols for “a” and “b” contacts [14]. (a) “a” contact, (b) “b” contact. Source: Based on IEEE [14].

Input. A physical quantity or quantities to which a relay is designed to respond.

Memory action. A method of retaining an effect of an input after the input ceases or is greatly reduced so that this (stored) input can still be used to produce the relay response.

Pickup. The action of a relay as it makes designated response to progressive increase of input. Also, the minimum value of an input quantity reached by progressive increase that will cause the relay to reach pickup state from reset.

Polarization. A term identifying the input that provides a reference for establishing the direction of system phenomena, such as the direction of power flow or direction to a fault or other disturbance.

Relay. An electric device designed to respond to input conditions in a prescribed manner and, after specified conditions are met, to respond to cause contact operation or similar change in associated electric control circuits.

Electromagnetic relay. An electromechanical relay magnet element, which is energized by the inputs quantity.

Static relay. A relay (or relay unit) in which the designated response is developed by electronic, solid state, magnetic, or other components without mechanical motion. A relay which is composed of both static and electromechanical units in which the designed response is accomplished by static units may be referred to as a static relay.

Digital relay or numeric relay is not yet defined in [3] or [10]. However, a digital or numeric relay is a subcategory of static relays. In this type of relay, analog measurements are digitized before being processed to produce the designed response. These devices are included in the broader classification of intelligent electronic device (IED).

IED. Any device incorporating one or more processors with the capability to receive or send data/control from or to an external source (e.g. electronic multifunction meters, digital relays, controllers) [3].

Reset. The action of a relay as it makes designated response to decrease in input. Reset, as a qualifying term, denotes the state of a relay when all response to decrease in input has been completed. Reset is also used to identify the maximum value of an input quantity reached by progressive decreases that will permit the relay to reach the state of complete reset from pickup. (Note: In defining the designated performance of relays having multiple inputs, reset describes the state when all inputs are zero and also when some input circuits are energized, if the resulting state is not altered from the zero-input condition.)

Electrically reset relay. A relay that is so constructed that it remains in the picked-up condition even after the input quantity is removed; an independent electric input is required to reset the relay.

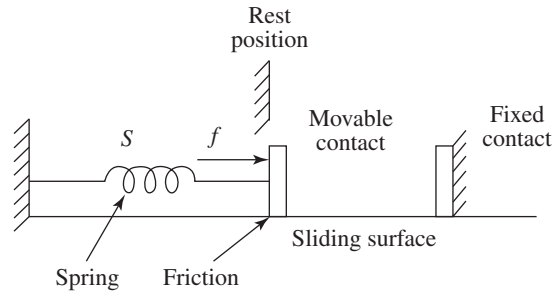
Hand reset relay. A relay that is so constructed that it remains in the picked up condition even after the input quantity is removed; specific manual reset action is required to reset the relay.

Self-reset relay (automatically reset relay). A relay that is so constructed that it returns to its reset position following an operation after the input quantity is removed.

The terms pickup, reset, and dropout are further described with the aid of the following discussion. Consider the two relay contacts shown in Figure A.2 and imagine that these contacts are part of an electromagnetic relay that is actuated by current to close a clapper on an electromagnetic relay circuit. Neglecting saturation, the force f acting on the member that holds the moving contact varies as the square of the current. The moving part is restrained from moving to the right by the spring "S." The contacts shown are "a" contacts since they are normally open when there is no current, hence no external force acting in the closing direction.

Now imagine that an input current flows, which is just sufficient to hold the contacts closed. This value of current is called the "pickup" current. A slight reduction in this input current will

Figure A.2 Diagram of the contacts of a simple electromechanical relay.



cause the contacts to part, but they do not return to the rest position since there is still sufficient current and friction to balance the force of the spring. Under this situation, we say that the relay has “dropped out.” If the current is not reduced such that the relay movable contact returns to the rest position, opening the contacts, we say that the relay has “reset,” and the value of current that just permits this to occur is the reset value. Reset action will occur, after a time delay, once the spring force exceeds the electromagnetic force of the current, plus any restraining forces such as friction, stiction, and other mechanical restraints.

(We note that, in magnetic devices that reduce an air gap when picking up, the current that will hold the relay in a picked-up state may be less than the pick-up current, since the air gap reduces during operation, requiring less current to produce the same force.)

Note that the rest position defined in Figure A.1 is not necessarily the zero current position. For example, it may be part of the relay design that a set of “b” contacts are attached that just open when the current exceeds that corresponding to the rest position. Any current below that which corresponds to the rest position current will cause the “b” contacts to remain closed.

Seal-in relay. An auxiliary relay that remains picked up through one of its own contacts, which bypasses the initiating circuit until deenergized by some other device.

Setting (noun). The desired characteristic, obtained as a result of having set a device, stated in terms of calibration markings or of actual performance bench marks, such as pickup current and operating time at a given value of input.

Trip-free relay. An auxiliary relay whose function is to open the closing circuit of an electrically operated switching device so that the opening operation can prevail over the closing operation.

The foregoing definitions are used repeatedly in the study of relays and protective systems in general. For additional definitions, the reader is referred to IEEE Standard 313 [10] and the IEEE Dictionary [3].

A.3 Classification of Relay Systems

Relays and relay systems may be classified by functions, by input, by operating principle, and by performance characteristics. The following tabulations give examples of each classification. The tabulations are not necessarily exhaustive, but are presented to illustrate the various forms of usage that have evolved in this discipline. We begin by presenting a classification of relays by functions, which are necessarily rather broad classifications. These classifications are presented in Table A.1.

A second classification is according to the input quantities to which the relay responds. The terms “over” and “under” are qualifying terms which may be used to prefix the input quantities, thereby

Table A.1 Classification of relays by function [10].

Function	Examples
Protective relay	See Tables A.2–A.4
Regulating relay	See Tables A.2–A.4
Monitoring relay	Alarm, fault detector, network phasing, synchronism check, verification
Programming relay	Accelerating, initiating, network master, phase selection, reclosing, synchronizing
Auxiliary relay	Blocking, closing, control, lockout, receiver, seal-in, timing, trip tree

Source: IEEE [10].

Table A.2 Classification of relays by input [10].

Current relay	Pressure relay
Flow relay	Temperature relay
Power relay	Voltage relay

Source: IEEE [10].

Table A.3 Classification of relays by operating principle or structure [10].^{a)}

Balance	Harmonic restraint	Quotient
Current balance	Latching	Replica temperature
Electrically reset	Mechanically reset	Restraint
Electromagnetic	Multirestraint	Self-reset
Electromechanical	Notching	Static
Gas accumulator	Percentage differential	Sudden pressure
Gas pressure	Phase sequence	Thermal
Hand reset	Product	

a) Add the word “relay” to each entry.

Source: IEEE [10].

classifying the relay on a performance basis. Other qualifying terms are: “ground,” “residual,” and “neutral.” Classifications according to input are given in Table A.2.

A third classification is according to operating principle or structure. A list of the members of this classification are given in Table A.3.

Table A.4 classifies relays by the relay performance characteristics. In Table A.4, the general performance prefix “over” indicates that the relay will operate for values of the input in excess of a predetermined threshold, while the prefix “under” indicates operation for input less than a given threshold. In some cases the relay operates for values both greater than and less than a predetermined value and are called over-and-under input relays.

Table A.4 Classification of relays by performance characteristics [10].^{a)}

Conductance	Mho	Positive sequence
Definite minimum time	Modified impedance	Power
Definite time	Negative sequence	Rate of change
Differential	Neutral	Reactance
Directional overcurrent	Open phase	Residual
Directional power	Overcurrent	Resistance
Directional	Overload	Reverse current
Distance	Overvoltage	Sector impedance
Frequency	Phase balance	Susceptance
Ground	Phase comparison	Time overcurrent
High speed	Phase	Undercurrent
Inverse time	Phase reversal	Undervoltage
Linear impedance	Phase undervoltage	Zero sequence

a) Add the word “relay” to each entry.

Source: IEEE [10].

Table A.5 Some general relay classifications.

Classification	Types
Speed	Slow speed High speed (50 ms or faster)
Intelligence source	Local Local plus remote Global
Computation	Analog Digital
Response designation	Static Mechanical action
Nature of response	Instantaneous Time delayed Definite time Inverse time versus current

In addition to the aforementioned, there are some general classifications that are often useful. These are given in Table A.5.

A brief study of the various relay classifications gives an indication of the extent of the effort that has been invested in this activity, both in protective device design and in the application of relay devices. This book is more concerned with applications, but it is obvious that some knowledge of available equipment is necessary.

A.4 Circuit Breaker Terms and Definitions

This is not a book with principal emphasis on circuit interruption or arc extinction. Nonetheless, these topics are important in the science of protection and deserve at least a brief consideration. A few definitions are presented here that will clarify the terminology related to circuit breakers and circuit interruption. Most of these terms are from ANSI Standards [15, 16], although the wording has been changed in some cases to make the definitions more readable in the present setting.

Circuit breaker. A circuit breaker is a device for closing, carrying, and interrupting a circuit between separable contacts under both load and fault conditions as prescribed in (C37.4) standards.

Circuit breakers are further defined in terms of the medium in which its contacting members and the circuit closing and interruption occur. This includes oil circuit breakers, air circuit breakers, compressed air circuit breakers, magnetic air circuit breakers, and oilless (other than oil) circuit breakers.

Circuit breakers are also classified in terms of voltage.

Operating voltage. The operating voltage of a circuit breaker is the rms line-to-line voltage of the system on which it is operated.

Recovery voltage. The recovery voltage is the voltage that occurs across the terminals of a pole of an ac circuit interrupting device upon interruption of current.

The phenomenon of interruption, or failure of the interruption process, is of considerable interest. Thus we define the following:

Restrike. A restrike is a resumption of current between the contacts of a circuit interrupting device during an opening operation after an interval of zero current of $1/4$ cycle at normal frequency or longer.

Reignition. Reignition is a resumption of current between the contacts of a circuit interrupting device during an opening operation after an interval of zero current of less than $1/4$ cycle at normal frequency.

It is well known that, when a fault occurs on a three-phase system, the currents in the three phases immediately following the application of the fault will exhibit dc offset in varying amounts, depending on the exact time of the fault with respect to the fundamental ac current wave. This requires the following definitions relating to circuit breaker currents.

Total short-circuit current (asymmetrical). The total short circuit current is the combination of the symmetrical component and the dc component, if any, of the short circuit current.

Symmetrical component (ac component). The symmetrical component of the short-circuit current is the normal-frequency ac component of the total current in rms amperes.

DC component. The dc component of the normal-frequency short-circuit current is that portion of the total short-circuit current that constitutes the asymmetry.

Degree of asymmetry. The degree of asymmetry of a current at any time is the ratio of the dc component to the peak value of the symmetrical component, determined from the envelope of the current wave at that time.

Making current. The making current of a circuit breaker, when closed on a short circuit, is the rms value of the total current, which is measured from the envelope of the current wave at the time of its first major peak.

Another group of definitions deals with construction details of the circuit breaker and its auxiliary control circuits. Since the relay system interacts directly with some of these auxiliary control circuits, these definitions are of direct interest.

Auxiliary switch. An auxiliary switch is a switch operated by the main switching device for signaling, interlocking, or other purposes.

Auxiliary switches in circuit breakers are classified as “a,” “b,” “aa,” “bb,” and “LC” for the purpose of specifying definite contact positions with respect to the main circuit breaker contacts. Thus, we have the following terms:

“a” auxiliary switch An “a” auxiliary switch is one that is open when the main switching device contacts are open.

As an aid to the memory, think of “a” meaning in “agreement” with the main circuit breaker contacts.

“b” auxiliary switch. A “b” auxiliary switch is one that is closed when the main switching device contacts are open.

As an aid to the memory, think of “b” as meaning “backwards” from the main circuit breaker contacts.

“aa” auxiliary switch. An “aa” auxiliary switch is one that is open when the main switching device operating mechanism is in the de-energized or non-operated position.

“bb” auxiliary switch. A “bb” auxiliary switch is one that is closed when the main switching device operating mechanism is in the de-energized or non-operated position.

“LC” auxiliary switch. An “LC” auxiliary switch is one that indicates the readiness of the mechanical linkage of the operating mechanism to close the breaker.

The abbreviation “LC” means “latch checking.” A latch checking auxiliary switch is closed when the circuit breaker mechanism linkage is relatched after an opening operation of the breaker.

Circuit breaker mechanism. A circuit breaker mechanism is the complete assembly of levers and other parts that actuates the movable contacts of the circuit breaker.

The circuit breaker is constructed so that it may be operated electrically to initiate either closing or opening of the circuit breaker switching device. This action is initiated by causing current to flow in coils provided for both closing and tripping.

Closing coil. A closing coil of a circuit breaker is a coil used in the electromagnet that supplies power for closing a circuit breaker.

Trip coil. A trip coil of a circuit breaker is a coil used in the electromagnet that initiates the opening of a circuit breaker.

It is also useful to consider certain functional definitions, such as the following:

Automatic tripping (automatic opening). Automatic tripping of a circuit breaker signifies the tripping of a circuit interrupter under predetermined or other conditions without the intervention of operating personnel; in other words, not *manual* tripping.

There are several types of automatic tripping that are of interest in system protection. The major types are the following:

Series overcurrent tripping. Series overcurrent tripping signifies the tripping of a circuit breaker from a trip coil in series with the main circuit, responsive to an increase in the main circuit current above a predetermined value.

Shunt tripping. Shunt tripping signifies the tripping of a circuit breaker by a trip coil energized from the same or a separate circuit or source of power; the trip coil circuit being closed through a relay, switch, or other means.

Transformer overcurrent tripping. Transformer overcurrent tripping signifies the tripping of a circuit breaker from a trip coil in series with the secondary windings of a current transformer whose primary winding is in series with the main circuit current above a predetermined value.

In addition to automatic tripping, the circuit breaker may be operated manually.

Non-automatic tripping Non-automatic tripping signifies the tripping of a circuit interrupter only in response to an act of operating personnel.

There are also several definitions regarding the term “trip free” as applied to circuit breakers. These definitions include the following:

Trip free. A circuit breaker is trip free when the tripping mechanism can trip the breaker even though the normal closing action is applied.

Mechanically trip free. A circuit breaker is trip free when the tripping mechanism can trip even though:

1. In a manually operated circuit breaker, the operating lever is being moved toward the closed position, or
2. In a power-operated circuit breaker, the operating mechanism is being moved toward the closed position either by continued application of closing power or by means of a maintenance closing lever.

Electrically trip free (anti-pump). An electrically operated circuit breaker is electrically trip free when the tripping mechanism can trip even though the closing control circuit is energized, and the closing mechanism will not reclose it after tripping until the closing control circuit is opened and again closed. However, the breaker may be held closed by the maintenance closing lever, unless it is also mechanically trip free.

It is important that circuit breakers be trip free for safety reasons. Suppose, for example, a breaker is positioned in a substation and protects a radial line. Suppose further that the line is undergoing maintenance and is permanently grounded by the maintenance personnel. Electrically, the line has a permanent three-phase fault to ground. Now, should an operator attempt to close the circuit breaker by manually energizing the closing coil, a trip-free breaker will trip the breaker even though the operator continues to hold the closing switch handle in the closed position.

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